Table 1-1 Fundamental SI units.

Dimension	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric charge	coulomb	C
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

special sections called *Technology Briefs* that are scattered throughout this book.

1-2 Dimensions, Units, and Notation

The ***International System of Units***, abbreviated ***SI*** after its French name *Système Internationale*, is the standard system used in today's scientific literature for expressing the units of physical quantities. Length is a ***dimension*** and meter is the ***unit*** by which it is expressed relative to a reference standard. The SI system is based on the units for the seven ***fundamental dimensions*** listed in **Table 1-1**. The units for all other dimensions are regarded as ***secondary*** because they are based on, and can be expressed in terms of, the seven fundamental units. Appendix A contains a list of quantities used in this book, together with their symbols and units.

For quantities ranging in value between 10^{-18} and 10^{18}, a set of prefixes arranged in steps of 10^3 are commonly used to denote multiples and submultiples of units. These prefixes, all of which were derived from Greek, Latin, Spanish, and Danish terms, are listed in **Table 1-2**. A length of 5×10^{-9} m, for example, may be written as 5 nm.

In EM, we work with scalar and vector quantities. In this book, we use a medium-weight italic font for symbols denoting scalar quantities, such as R for resistance, and a boldface roman font for symbols denoting vectors, such as $\mathbf{E}$ for the electric field vector. A vector consists of a magnitude (scalar) and a direction, with the direction usually denoted by a unit vector. For example,

$$\mathbf{E} = \hat{\mathbf{x}}E, \tag{1.1}$$

where E is the magnitude of $\mathbf{E}$ and $\hat{\mathbf{x}}$ is its direction. A symbol denoting a unit vector is printed in boldface with a circumflex (ˆ) above it.

Throughout this book, we make extensive use of ***phasor representation*** in solving problems involving electromagnetic quantities that vary sinusoidally in time. Letters denoting phasor quantities are printed with a tilde (∼) over the letter. Thus, $\widetilde{\mathbf{E}}$ is the phasor electric field vector corresponding to the instantaneous electric field vector $\mathbf{E}(t)$. This notation is discussed in more detail in Section 1-7.

Table 1-2 Multiple and submultiple prefixes.

Prefix	Symbol	Magnitude
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}

Notation Summary

- **Scalar quantity:** medium-weight italic, such as C for capacitance.
- **Units:** medium-weight roman, as in V/m for volts per meter.
- **Vector quantities:** boldface roman, such as $\mathbf{E}$ for electric field vector
- **Unit vectors:** boldface roman with circumflex (ˆ) over the letter, as in $\hat{\mathbf{x}}$.
- **Phasors:** a tilde (∼) over the letter; $\widetilde{E}$ is the phasor counterpart of the sinusoidally time-varying scalar field $E(t)$, and $\widetilde{\mathbf{E}}$ is the phasor counterpart of the sinusoidally time-varying vector field $\mathbf{E}(t)$.

1-3 The Nature of Electromagnetism

Our physical universe is governed by four fundamental forces of nature:

- The ***nuclear force***, which is the strongest of the four, but its range is limited to ***subatomic scales***, such as nuclei.
- The ***electromagnetic force*** exists between all charged

1-2 量纲、单位与符号

（承第3页历史沿革）这些技术将在散布于本书各处的“技术简介”专栏中介绍。

国际单位制（International System of Units，SI）是科学文献中表达物理量单位的标准单位制。长度是一种量纲，米是将长度与参考标准比较时采用的单位。其他物理量的单位可由基本单位组合得到，称为导出单位。附录A列出了本书使用的物理量、符号及单位。

表1-1 原书所列的基本单位

物理量	单位	符号
长度	米	m
质量	千克	kg
时间	秒	s
电荷量〔见校注〕	库仑	C
热力学温度	开尔文	K
物质的量	摩尔	mol
发光强度	坎德拉	cd

校注：原书将电荷量及库仑列入“基本SI单位”，这一表述不符合正式SI分类。SI的电学基本量是电流，基本单位是安培（A）；库仑是导出单位，1 C = 1 A·s。本表保留原书行项供对照。依据：国际计量局（BIPM）SI base units，https://www.bipm.org/en/measurement-units/si-base-units 。

对10^{-18}至10^{18}范围内的量，常采用按10^3倍率递进的词头表示单位的十进倍数或分数。这些词头源自希腊语、拉丁语、西班牙语和丹麦语；例如5 × 10^{-9} m可写为5 nm。

表1-2 原书所列SI词头

词头（英文）	符号	倍率	词头（英文）	符号	倍率
艾（exa）	E	10^{18}	毫（milli）	m	10^{-3}
拍（peta）	P	10^{15}	微（micro）	μ	10^{-6}
太（tera）	T	10^{12}	纳（nano）	n	10^{-9}
吉（giga）	G	10^{9}	皮（pico）	p	10^{-12}
兆（mega）	M	10^{6}	飞（femto）	f	10^{-15}
千（kilo）	k	10^{3}	阿（atto）	a	10^{-18}

电磁学涉及标量与矢量。本书用普通字重的斜体表示标量（如电阻R），用粗体正体表示矢量（如电场强度**E**）。矢量具有大小和方向，其方向通常由单位矢量表示。例如：

$$\mathbf{E} = \hat{\mathbf{x}}E, \tag{1.1}$$

式（1.1）中，标量E表示矢量**E**的大小，带帽号的粗体**x**表示沿x方向的单位矢量。本书用相量处理随时间作正弦变化的电磁量：在相应标量或矢量符号上加波浪号（~），表示其相量；瞬时量仍写为E(t)或**E**(t)。详见第1-7节。

符号约定：标量用普通斜体；单位用普通正体（如V/m）；矢量用粗体正体；单位矢量在粗体字母上加帽号；相量在相应符号上加波浪号。须保留这些区别，不能将相量与瞬时量、标量与矢量混同。

1-3 电磁学的本质

自然界存在四种基本相互作用。核力（此处指强相互作用）最强，但作用范围局限于原子核等亚原子尺度。电磁力存在于带电粒子之间。（接下页）

particles. It is the dominant force in ***microscopic*** systems, such as atoms and molecules, and its strength is on the order of 10^{-2} that of the nuclear force.

- The ***weak-interaction force***, whose strength is only 10^{-14} that of the nuclear force. Its primary role is in interactions involving certain radioactive elementary particles.
- The ***gravitational force*** is the weakest of all four forces, having a strength on the order of 10^{-41} of the nuclear force. However, it often is the dominant force in ***macroscopic*** systems, such as the solar system.

This book focuses on the electromagnetic force and its consequences. Even though the electromagnetic force operates at the atomic scale, its effects can be transmitted in the form of electromagnetic waves that can propagate through both free space and material media. The purpose of this section is to provide an overview of the basic ***framework of electromagnetism***, which consists of certain fundamental laws governing the electric and magnetic fields induced by static and moving electric charges, the relations between the electric and magnetic fields, and how these fields interact with matter. As a precursor, however, we will take advantage of our familiarity with gravitational force by describing some of its properties because they provide a useful analogue to those of electromagnetic force.

1-3.1 Gravitational Force: A Useful Analogue

According to Newton's law of gravity, the gravitational force $\mathbf{F}_{g_{21}}$ acting on mass m_2 due to a mass m_1 at a distance R_{12} from m_2 (**Fig. 1-3**) is given by

$$\mathbf{F}_{g_{21}} = -\hat{\mathbf{R}}_{12}\frac{Gm_1m_2}{R_{12}^2} \quad \text{(N)}, \tag{1.2}$$

where G is the universal gravitational constant, $\hat{\mathbf{R}}_{12}$ *is a unit vector that points from* m_1 *to* m_2, and the unit for force is newton (N). The negative sign in Eq. (1.2) accounts for the fact that the gravitational force is attractive. Conversely, $\mathbf{F}_{g_{12}} = -\mathbf{F}_{g_{21}}$, where $\mathbf{F}_{g_{12}}$ is the force acting on mass m_1 due to the gravitational pull of mass m_2. *Note that the first subscript of* $\mathbf{F}_g$ *denotes the mass experiencing the force and the second subscript denotes the source of the force.*

► The force of gravitation acts at a distance. ◄

The two objects do not have to be in direct contact for each to experience the pull by the other. This phenomenon of action at a distance has led to the concept of ***fields***. An object of mass m_1 induces a ***gravitational field*** $\boldsymbol{\psi}_1$ (**Fig. 1-4**) that does not physically emanate from the object, yet its influence exists at every point in space such that, if another object of mass m_2 were to exist at a distance R_{12} from the object of mass m_1, the object of mass m_2 would experience a force acting on it equal to

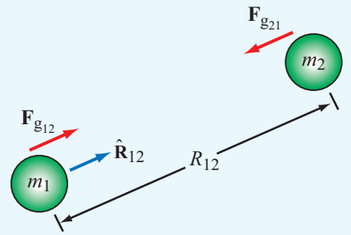


Figure 1-3 Gravitational forces between two masses.

$$\mathbf{F}_{g_{21}} = \boldsymbol{\psi}_1 m_2, \tag{1.3}$$

where

$$\boldsymbol{\psi}_1 = -\hat{\mathbf{R}}\frac{Gm_1}{R^2} \quad \text{(N/kg)}. \tag{1.4}$$

In Eq. (1.4), $\hat{\mathbf{R}}$ is a unit vector that points in the radial direction away from object m_1; therefore, $-\hat{\mathbf{R}}$ points toward m_1. The force due to $\boldsymbol{\psi}_1$ acting on a mass m_2, for example, is obtained from the combination of Eqs. (1.3) and (1.4) with $R = R_{12}$ and $\hat{\mathbf{R}} = \hat{\mathbf{R}}_{12}$. The field concept may be generalized by defining the gravitational field $\boldsymbol{\psi}$ at any point in space such that, when a test mass m is placed at that point, the force $\mathbf{F}_g$ acting on it is related to $\boldsymbol{\psi}$ by

$$\boldsymbol{\psi} = \frac{\mathbf{F}_g}{m}. \tag{1.5}$$

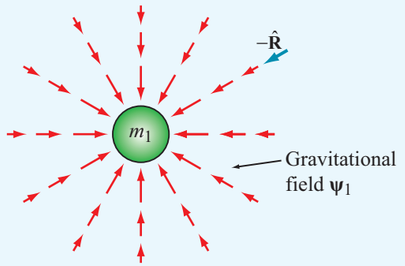


Figure 1-4 Gravitational field $\boldsymbol{\psi}_1$ induced by a mass m_1.

1-3.1 引力：一种有用的类比

（承上页）电磁力在原子、分子等微观系统中占主导地位，按原书所用的数量级比较，其强度约为核力的10^{-2}。弱相互作用力约为核力的10^{-14}，主要涉及某些放射性基本粒子的相互作用。引力最弱，约为核力的10^{-41}，却常在太阳系等宏观系统中占主导地位。

本书讨论电磁力及其效应。尽管电磁力作用于原子尺度，其效应却能通过在自由空间和物质介质中传播的电磁波传递。本节概述电磁学的基本框架：静止和运动电荷所产生的电场、磁场遵循哪些基本定律，电场与磁场有何联系，以及它们如何与物质相互作用。首先介绍熟悉的引力，借此类比电磁力的性质。

根据牛顿万有引力定律，质量为m_1的物体对相距R_{12}、质量为m_2的物体施加的引力$\mathbf{F}_{g21}$为：

$$\mathbf{F}_{g_{21}} = -\hat{\mathbf{R}}_{12}\frac{Gm_1m_2}{R_{12}^2} \quad \text{(N)}, \tag{1.2}$$

G为万有引力常量，R_{12}上方的帽号表示由m_1指向m_2的单位矢量，力的单位是牛顿（N）。负号表示引力为吸引力。反过来，m_2对m_1的作用力满足$\mathbf{F}_{g12} = -\mathbf{F}_{g21}$。力符号的第一个下标表示受力物体，第二个下标表示施力物体。

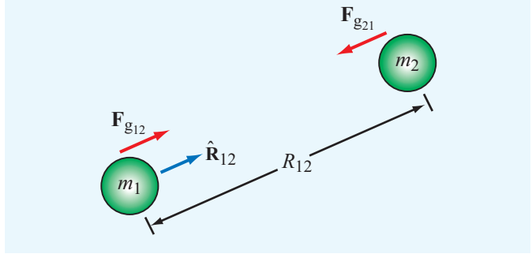


图1-3 两个有质量物体之间的引力。

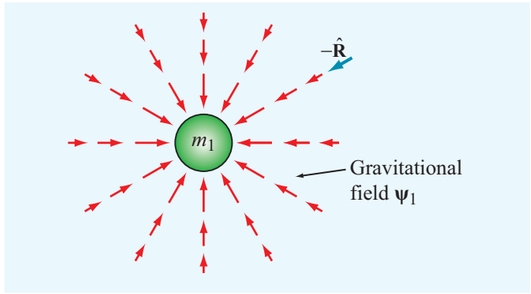


图1-4 质量m_1产生的引力场ψ_1。

引力可以在一定距离外起作用，两个物体无须直接接触。这种超距作用现象引出了“场”的概念。质量m_1产生引力场$\boldsymbol{\psi}_1$；场并非从物体中发射出来的某种物质，但其影响遍及空间。位于该场中的质量m_2受到：

$$\mathbf{F}_{g_{21}} = \boldsymbol{\psi}_1 m_2, \tag{1.3}$$

$$\boldsymbol{\psi}_1 = -\hat{\mathbf{R}}\frac{Gm_1}{R^2} \quad \text{(N/kg)}. \tag{1.4}$$

式（1.4）中，带帽号的$\mathbf{R}$沿径向背离m_1，其负方向指向m_1。在式（1.3）、（1.4）中令$R = R_{12}$，并令径向单位矢量取m_1指向m_2的方向，即得m_2所受的力。一般地，在空间某点放置试验质量m，该点的引力场定义为单位质量所受的引力：

$$\boldsymbol{\psi} = \frac{\mathbf{F}_g}{m}. \tag{1.5}$$

图中Gravitational field意为“引力场”。本页的ψ为矢量，不能省略方向信息。原书四种相互作用的强度比仅作其选定尺度下的概略比较，并非与能标无关的普适常数。

The force $\mathbf{F}_g$ may be due to a single mass or a collection of many masses.

1-3.2 Electric Fields

The electromagnetic force consists of an electrical component $\mathbf{F}_e$ and a magnetic component $\mathbf{F}_m$.

► The electrical force $\mathbf{F}_e$ is similar to the gravitational force, but with two major differences:

(1) the source of the electrical field is electric charge, not mass, and

(2) even though both types of fields vary inversely as the square of the distance from their respective sources, electric charges may have positive or negative polarity, resulting in a force that may be attractive or repulsive. ◄

We know from atomic physics that all matter contains a mixture of neutrons, positively charged protons, and negatively charged electrons with the fundamental quantity of charge being that of a single electron, usually denoted by the letter e. The unit by which electric charge is measured is the coulomb (C), named in honor of the eighteenth-century French scientist Charles Augustin de Coulomb (1736–1806). The magnitude of e is

$$e = 1.6 \times 10^{-19} \quad \text{(C)}. \tag{1.6}$$

The charge of a single electron is $q_e = -e$ and that of a proton is equal in magnitude but opposite in polarity: $q_p = e$.

► Coulomb's experiments demonstrated that:

(1) two like charges repel one another, whereas two charges of opposite polarity attract,

(2) the force acts along the line joining the charges, and

(3) its strength is proportional to the product of the magnitudes of the two charges and inversely proportional to the square of the distance between them. ◄

These properties constitute what today is called ***Coulomb's law***, which can be expressed mathematically as

$$\mathbf{F}_{e_{21}} = \hat{\mathbf{R}}_{12} \frac{q_1 q_2}{4\pi\varepsilon_0 R_{12}^2} \quad \text{(N)} \quad \textbf{(in free space)}, \tag{1.7}$$

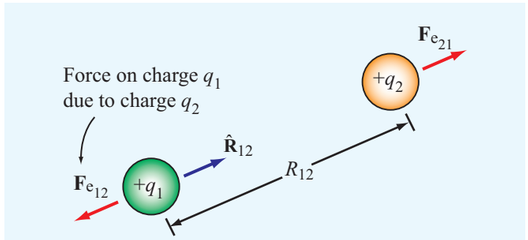


Figure 1-5 Electric forces on two positive point charges in free space.

where $\mathbf{F}_{e_{21}}$ is the ***electrical force*** acting on charge q_2 due to charge q_1 when both are in ***free space*** (vacuum), R_{12} is the distance between the two charges, $\hat{\mathbf{R}}_{12}$ *is a unit vector pointing from charge q_1 to charge q_2* (**Fig. 1-5**), and ε_0 is a universal constant called the ***electrical permittivity of free space*** [$\varepsilon_0 = 8.854 \times 10^{-12}$ farad per meter (F/m)]. The two charges are assumed to be isolated from all other charges. The force $\mathbf{F}_{e_{12}}$ acting on charge q_1 due to charge q_2 is equal to force $\mathbf{F}_{e_{21}}$ in magnitude, but opposite in direction: $\mathbf{F}_{e_{12}} = -\mathbf{F}_{e_{21}}$.

The expression given by Eq. (1.7) for the electrical force is analogous to that given by Eq. (1.2) for the gravitational force, and we can extend the analogy further by defining the existence of an ***electric field intensity*** $\mathbf{E}$ due to any charge q as

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon_0 R^2} \quad \text{(V/m)} \quad \textbf{(in free space)}, \tag{1.8}$$

where R is the distance between the charge and the observation point, and $\hat{\mathbf{R}}$ *is the radial unit vector pointing away from the charge.* **Figure 1-6** depicts the electric field lines due to a positive charge. For reasons that will become apparent in later chapters, the unit for $\mathbf{E}$ is volt per meter (V/m).

► If a point charge q' is present in an electric field $\mathbf{E}$ (due to other charges), the point charge will experience a force acting on it equal to $\mathbf{F}_e = q'\mathbf{E}$. ◄

Electric charge exhibits two important properties.

► The first property of electric charge is encapsulated by the ***law of conservation of electric charge***, which states that *the (net) electric charge can neither be created nor destroyed*. ◄

1-3.2 电场

（承上页）引力$\mathbf{F}_g$可以由单个物体产生，也可以由多个物体共同产生。电磁力由电场力$\mathbf{F}_e$与磁场力$\mathbf{F}_m$组成。电场力与引力类似，但有两点主要区别：电场的源是电荷而非质量；虽然孤立点源的场强均随距离平方的倒数变化，电荷却有正、负之分，因而电场力可以是吸引力，也可以是排斥力。

原子物理学表明，普通物质包含中子、带正电的质子和带负电的电子。元电荷e是单个电子所带电荷量的绝对值。电荷量的单位库仑（C），得名于法国科学家Charles Augustin de Coulomb（1736—1806）。原书采用：

$$e = 1.6 \times 10^{-19} \quad \text{(C)}. \tag{1.6}$$

电子的电荷量为$q_e = -e$，质子的电荷量为$q_p = e$，二者大小相等、符号相反。

库仑实验表明：（1）同号电荷相斥，异号电荷相吸；（2）力沿两电荷的连线方向作用；（3）力的大小与两电荷量绝对值的乘积成正比，与间距的平方成反比。以上性质构成库仑定律：

$$\mathbf{F}_{e_{21}} = \hat{\mathbf{R}}_{12} \frac{q_1 q_2}{4\pi\varepsilon_0 R_{12}^2} \quad \text{(N)} \quad \textbf{(in free space)}, \tag{1.7}$$

式（1.7）适用于自由空间（真空）中与其他电荷隔离的两个点电荷。$\mathbf{F}_{e21}$为q_1作用于q_2的电场力；R_{12}为间距，带帽号的$\mathbf{R}_{12}$由q_1指向q_2。ε_0为真空介电常数，约为8.854×10^{-12} F/m。反向作用力满足$\mathbf{F}_{e12} = -\mathbf{F}_{e21}$。

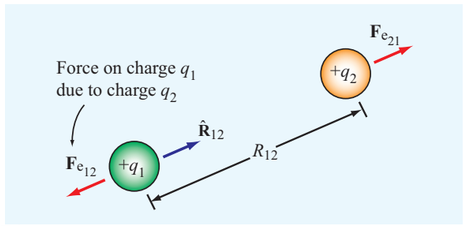


图1-5 自由空间中两个正点电荷所受的电场力。

式（1.7）的电场力表达式与式（1.2）的引力表达式相似。进一步类比，可将孤立点电荷q在自由空间中产生的电场强度定义为：

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon_0 R^2} \quad \text{(V/m)} \quad \textbf{(in free space)}, \tag{1.8}$$

R是电荷与观察点的距离，带帽号的$\mathbf{R}$是由电荷指向观察点的径向单位矢量。图1-6给出正电荷的电场线。电场强度$\mathbf{E}$的单位为伏特每米（V/m）。若点电荷q′处于其他电荷产生的外电场$\mathbf{E}$中，则所受电场力为$\mathbf{F}_e = q'\mathbf{E}$。

电荷具有两项重要性质。第一项是电荷守恒：净电荷既不能凭空产生，也不能消失。

图中Force on charge q_1 due to charge q_2：q_2对q_1施加的力。式（1.7）、（1.8）旁的in free space均指“在自由空间中”。

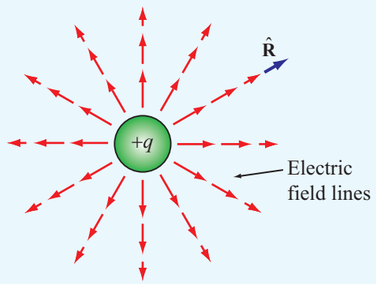


Figure 1-6 Electric field **E** due to charge q.

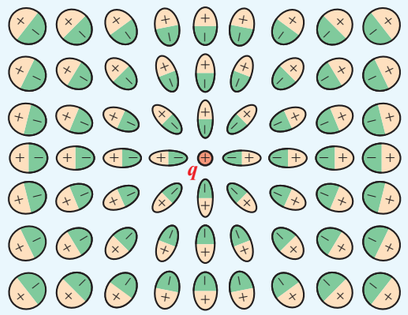


Figure 1-7 Polarization of the atoms of a dielectric material by a positive charge q.

If a volume contains n_p protons and n_e electrons, then its total charge is

$$q = n_p e - n_e e = (n_p - n_e)e \quad \text{(C)}. \tag{1.9}$$

Even if some of the protons were to combine with an equal number of electrons to produce neutrons or other elementary particles, the net charge q remains unchanged. In matter, the quantum mechanical laws governing the behavior of the protons inside the atom's nucleus and the electrons outside it do not allow them to combine.

► The second important property of electric charge is embodied by the ***principle of linear superposition***, which states that *the total vector electric field at a point in space due to a system of point charges is equal to the vector sum of the electric fields at that point due to the individual charges*. ◄

This seemingly simple concept allows us in future chapters to compute the electric field due to complex distributions of charge without having to be concerned with the forces acting on each individual charge due to the fields by all of the other charges.

The expression given by Eq. (1.8) describes the field induced by an electric charge residing in free space. Let us now consider what happens when we place a positive point charge in a material composed of atoms. In the absence of the point charge, the material is electrically neutral with each atom having a positively charged nucleus surrounded by a cloud of electrons of equal but opposite polarity. Hence, at any point in the material not occupied by an atom, the electric field **E** is zero. Upon placing a point charge in the material, as shown in **Fig. 1-7**, the atoms experience forces that cause them to become distorted. The center of symmetry of the electron cloud is altered with respect to the nucleus with one pole of the atom becoming positively charged relative to the other pole. Such a polarized atom is called an ***electric dipole***, and the distortion process is called ***polarization***. The degree of polarization depends on the distance between the atom and the isolated point charge, and the orientation of the dipole is such that the axis connecting its two poles is directed toward the point charge, as illustrated schematically in **Fig. 1-7**. The net result of this polarization process is that the electric fields of the dipoles of the atoms (or molecules) tend to counteract the field due to the point charge. Consequently, the electric field at any point in the material is different from the field that would have been induced by the point charge in the absence of the material. To extend Eq. (1.8) from the free-space case to any medium, we replace the permittivity of free space ε_0 with ε, where ε is the permittivity of the material in which the electric field is measured and is therefore characteristic of that particular material. Thus,

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{1.10}$$

(material with permittivity ε)

Often, ε is expressed in the form

$$\varepsilon = \varepsilon_r \varepsilon_0 \quad \text{(F/m)}, \tag{1.11}$$

where ε_r is a dimensionless quantity called the ***relative permittivity*** or ***dielectric constant*** of the material. For a vacuum, $\varepsilon_r = 1$; for air near the Earth's surface, $\varepsilon_r = 1.0006$; and the values of ε_r for materials that we have occasion to use in this book are tabulated in Appendix B.

电荷叠加与介质极化

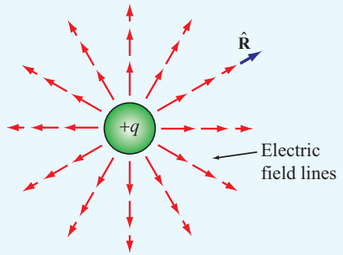


图1-6 点电荷q产生的电场E。

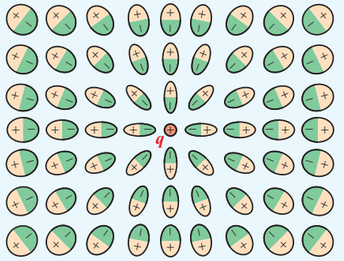


图1-7 正点电荷q使介质原子极化。

如果某体积内有n_p个质子和n_e个电子，其总电荷量为：

$$q = n_p e - n_e e = (n_p - n_e)e \quad \text{(C)}. \tag{1.9}$$

即便若干质子与等量电子结合形成中子或其他基本粒子，净电荷量q也保持不变。原书接着指出：在通常的物质中，控制核内质子与核外电子行为的量子规律使原子保持稳定，而不会简单地因正、负电荷相吸就全部结合。

电荷的第二项重要性质体现在线性叠加原理中：多个点电荷在空间某点产生的总电场，等于各个点电荷单独在该点产生的电场的矢量和。这使我们能根据给定的电荷分布计算电场，无须先求出其他电荷对每个电荷的作用力。

式（1.8）给出自由空间中点电荷产生的场。现在考虑把正点电荷放入由原子组成的介质中。加入该点电荷前，介质整体电中性；每个原子的正电核由带等量负电荷的电子云包围。原书在此采用未极化介质的简化模型，认为无外加电荷时其宏观电场为零。

放入正点电荷后，原子的电子云相对于原子核发生偏移或畸变，使原子一侧相对带正电，另一侧相对带负电。这种极化的原子称为电偶极子，该过程称为极化。极化程度与原子到点电荷的距离有关；偶极子的轴线沿原子与点电荷的连线排列，如图1-7所示。

原子或分子偶极子产生的电场总体上趋于削弱该点电荷原有的电场。因此，介质中的场与不存在介质时不同。原书把式（1.8）中的真空介电常数ε_0换为介质的介电常数ε，得到：

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{1.10}$$

(material with permittivity ε)

常将介电常数写为：

$$\varepsilon = \varepsilon_r \varepsilon_0 \quad \text{(F/m)}, \tag{1.11}$$

ε_r为无量纲的相对介电常数，英文也称dielectric constant。真空中$\varepsilon_r = 1$；原书给出近地面空气的近似值1.0006。其他常用材料的取值见附录B。

适用条件校注：式（1.10）直接适用于充满空间的均匀、各向同性线性介质中的孤立点电荷，不能不加条件地用于任意非均匀介质、各向异性介质或有界面的问题。“未极化时宏观场为零”也不等于原子间任意微观位置的场严格为零。图中文字Electric field lines：电场线。

In addition to the electric field intensity **E**, we often find it convenient to also use a related quantity called the ***electric flux density*** **D** given by

$$\mathbf{D} = \varepsilon\mathbf{E} \qquad (\mathrm{C/m^2}), \tag{1.12}$$

with unit of coulomb per square meter ($\mathrm{C/m^2}$).

► These two electric quantities, **E** and **D**, constitute one of two fundamental pairs of electromagnetic fields. The second pair consists of the magnetic fields discussed next. ◄

1-3.3 Magnetic Fields

As early as 800 BC, the Greeks discovered that certain kinds of stones exhibit a force that attracts pieces of iron. These stones are now called ***magnetite*** (Fe_3O_4), and the phenomenon they exhibit is known as ***magnetism***. In the thirteenth century, French scientists discovered that, when a needle was placed on the surface of a spherical natural magnet, the needle oriented itself along different directions for different locations on the magnet. By mapping the directions indicated by the needle, it was determined that the magnetic force formed magnetic field lines that encircled the sphere and appeared to pass through two points diametrically opposite to each other. These points, called the ***north and south poles*** of the magnet, were found to exist for every magnet, regardless of its shape. The magnetic-field pattern of a bar magnet is displayed in **Fig. 1-8**. It was also observed that like poles of different magnets repel each other and unlike poles attract each other.

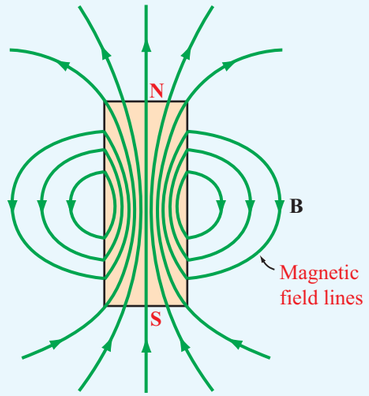


Figure 1-8 Pattern of magnetic field lines around a bar magnet.

► The attraction–repulsion property for magnets is similar to the electric force between electric charges, except for one important difference: ***Electric charges can be isolated, but magnetic poles always exist in pairs.*** ◄

If a permanent magnet is cut into small pieces, no matter how small each piece is, it will always have a north and a south pole.

The magnetic lines surrounding a magnet represent the ***magnetic flux density*** **B**. A magnetic field not only exists around permanent magnets but also can be created by electric current. This connection between electricity and magnetism was discovered in 1819 by the Danish scientist Hans Oersted (1777–1851), who observed that an electric current in a wire caused a compass needle placed in its vicinity to deflect and that the needle turned so that its direction was always perpendicular to the wire and to the radial line connecting the wire to the needle. From these observations, he deduced that the current-carrying wire induced a magnetic field that formed closed circular loops around the wire (**Fig. 1-9**). Shortly after Oersted's discovery, French scientists Jean-Baptiste Biot and Felix Savart developed an expression that relates the magnetic flux density **B** at a point in space to the current I in the conductor. Application of their formulation, known today as the ***Biot–Savart law***, to the situation depicted in **Fig. 1-9** for

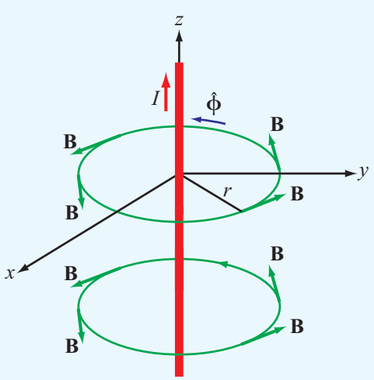


Figure 1-9 The magnetic field induced by a steady current flowing in the z direction.

1-3.3 磁场

除电场强度**E**外，还常使用与之相关的电通量密度（电位移）**D**：

$$\mathbf{D} = \varepsilon\mathbf{E} \qquad (\mathrm{C/m^2}), \tag{1.12}$$

其单位为库仑每平方米（$\mathrm{C/m^2}$）。**E**与**D**构成两对基本电磁场量中的一对，另一对为下文讨论的磁场量。

早在公元前800年，希腊人就发现某些石头能吸引铁块。它们现称磁铁矿（Fe_3O_4），这种现象称为磁性。13世纪，法国科学家把磁针置于球形天然磁体表面的不同位置，观察到磁针的指向随位置改变。连接这些方向，可绘出环绕球体、并通过一对径向相对点的磁场线。这两点称为磁北极和磁南极。无论磁体形状如何，都有南、北两极。条形磁体的磁场线见图1-8。实验还表明，同名磁极相斥，异名磁极相吸。

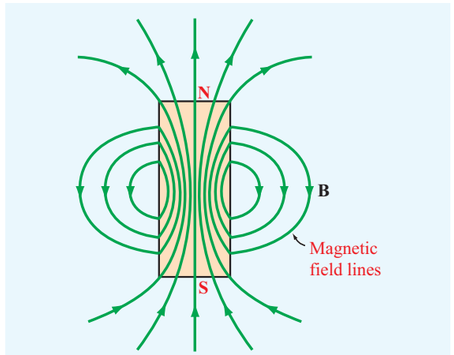


图1-8 条形磁体周围的磁场线。

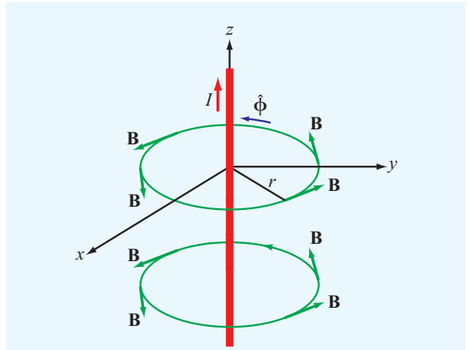


图1-9 沿z方向的恒定电流产生的磁场。

磁体的吸斥性质与电荷之间的作用类似，但有一个重要区别：正、负电荷可以单独存在，而通常磁体的磁极总是成对出现。把永磁体切成任意小块，每块仍有南、北两极。

磁体周围的磁场线表示磁通量密度**B**。磁场既存在于永磁体周围，也能由电流产生。原书记载，丹麦科学家Hans Oersted（1777—1851）于1819年发现电流会使附近的罗盘磁针偏转；磁针最终同时垂直于导线和导线至磁针的径向连线。

据此，他推断载流直导线周围的磁场线是环绕导线的闭合圆，如图1-9。随后，法国科学家Jean-Baptiste Biot与Felix Savart建立了空间某点磁通量密度**B**与导体电流I之间的关系，称为毕奥-萨伐尔定律。将其用于自由空间中的无限长直导线，即得到下页的式（1.13）。

图中Magnetic field lines：磁场线。图1-9的r为至导线的径向距离，带帽号的φ为方位角方向单位矢量。这里的电场与磁场本构关系均以原书采用的线性、各向同性介质模型为前提。